YOUSSOF SOLIMAN

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EDUCATION

Colorado School of Mines, Golden, CO — May 2028
B.S. in Electrical Engineering | GPA: 3.938 | Pursuing a M.S. in Integrated Circuits
Relevant Courses: Digital Logic, Circuits

EXPERIENCE

Mines Formula SAE, *Electrical Engineer*, Golden, CO — Aug 2025 – Present

- Designed low voltage electronics for an EV/IC race car capable of 0-60 in under 4 seconds.
- Responsible engineer for multiple hardware projects, including a 12-channel ADC analog board and an advanced STM32 development board, to support sensor data acquisition and software testing across the team.
- Executed post-fabrication board bring-up and hardware validation, identifying critical hardware issues prior to integration.

ARC Lab, *Undergraduate Researcher*, Golden, CO — Aug 2026 – Present

- Modeled high-frequency signals to analyze wave propagation and signal integrity characteristics.
- Applied Ansys HFSS to accurately simulate and evaluate complex electromagnetic effects in passive components.

Colorado School of Mines, *Electrical Engineering Lab Technician*, Golden, CO — Aug 2026 – Present

- Maintained bench calibration and lab readiness across multiple EE instructional labs to ensure safety compliance.

PROJECTS

ADC Analog Board - Formula SAE:

- Architected a 12-channel 16-bit mixed-signal data-acquisition board around an STM32 MCU and a precision ADC to collect 12 linear potentiometer inputs from various vehicle sensors.
- Integrated Ethernet (RMII), CAN, SDMMC1, I2C, and SPI for sensor communication and telemetry while maintaining signal integrity, with a 34-pin AMPSEAL connector selected for harness integration with the rest of the car.
- Engineered mixed-signal circuit board layouts prioritizing signal integrity for high-speed Ethernet (RMII) and CAN interfaces, utilizing impedance matching and trace-length tuning to reduce EMI and crosstalk.
- Implemented power design centered around a 4-rail LDO and buck-converter power tree (2 digital, 2 analog).

Mini-Oscilloscope:

- Independently designed a 4-layer oscilloscope PCB in Altium around an STM32, managing the full design cycle from schematic capture to layout design.
- Design features included USB signal routing, power supply, op-amps, analog front-end debugging for signal integrity, and GND/PWR plane stitching.
- Ran design rule checks (DRCs) and electrical rule checks (ERCs) before fabricating through JLCPCB and confirmed correct power delivery and overall board functionality. Embedded firmware in C++ through STM32CubeIDE.
- Executed comprehensive post-fabrication board bring-up and hardware validation, diagnosing and resolving a critical I2C bus hang to restore booting.

Advanced STM32 Development Board - Formula SAE:

- Designed and brought up a functional, 100+ pin STM32 development board, enabling the software team to run tests and actively debug the car.
- Circuit design included EMI mitigation, integrating CAN transceivers, I2C, SPI, JTAG, and Arduino GPIOs for Uno integration.
- Designed the board's power tree (buck converter, LDO, USB-C power delivery).

Thermocouple Reading Device V1 (TRD-1) - Formula SAE:

- Designed and produced a temperature sensor using Altium for a high temperature (200C) custom oven for baking race car carbon fiber parts.
- Replaced the aerodynamics team's improvised multimeter setups with a dedicated thermal monitoring solution.

SKILLS

- **Technical Skills:** Soldering, Oscilloscopes, Signal Generators, Altium Designer, PCB Bring-up, High-speed signal routing, Board validation, Power design, Diodes, FETs, I2C, SPI, UART, Ethernet, USB, JTAG, Ansys, LTSpice, STM32CubeMX/IDE, SolidWorks (CSWA), Arduino, MATLAB
- **Computer Skills:** C/C++, C#, Python, Git, Linux/UNIX, Microsoft Office Products
- **Professional Skills:** Interdisciplinary collaboration, Project management